JVET 39차 표준화 회의 출장 보고서

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1. 출장 개요

1.1. 출장자

- 소속: 동아대학교 컴퓨터 AI 공학부 영상신호처리연구실
- 성명: 김선재/이태식 박사과정, 박병주/이동호/정현아 석사과정

1.2. 출장지 및 기관

- 대한민국 대전 / Daejeon Convention Center, DCC

1.3. 출장 목적

- JVET 39 차, MPEG 151 차 국제 표준화 회의 참석

1.4. 출장 기간

- **표준화 회의 기간 (8 박 9 일): 2025 년 06 월 26 일 (목) ~ 2025 년 07 월 04 일 (금)**
- 김선재/이태식/정현아
  - 총 출장 기간 (8 박 9 일): 2025 년 06 월 25 일 (수) ~ 2025 년 07 월 03 일 (목)
- 박병주
  - 총 출장 기간 (6 박 7 일): 2025 년 06 월 25 일 (수) ~ 2025 년 07 월 01 일 (화)
- 이동호
  - 총 출장 기간 (5 박 6 일): 2025 년 06 월 28 일 (토) ~ 2025 년 07 월 03 일 (목)

1.5. 관련 과제 및 수행 업무

생략 (공개용 보고서)

1.6. 수행과제와의 연계성(출장목적)

생략 (공개용 보고서)

✓

1.7. 세부 일정 및 활동 내용

일 시	활동 내용
2025.06.25 (수) ~ 2025.06.27 (금)	- 김선재/이태식 박사과정, 박병주/정현아 석사과정 • 부산역 KTX → 대전역 KTX: 06/25(수) 16:15 ~ 17:55 (1 시간 40 분) - 이동호 석사과정 • 서울역 KTX → 대전역 KTX: 06/27(금) 20:09 ~ 21:31 (1 시간 43 분)
2025.06.26 (목)	Thu. 26 June, 1st day

~ 2025.07.03 (목)	• 0910–1000 Opening remarks, review of practices, agenda, IPR policy reminder • 1000–1320 Reports of AHGs 1-18 • 1500–1620 Review of EE1 summary report • 1650–1950 Review of EE2 summary report • 1500–2000 HLS 6.1, 6.2 Fri. 27 June, 2nd day • 0800–1230 Planning CfE dry run, review of EE2 summary report, EE2 related and non-EE • 0900–1300 HLS 6.1, 6.2.3, 6.2.4, 6.2.5, 6.2.6 (Rm. 202, chaired by Jill Boyce) • 1330–2000 Viewing for CfE dry-run (no low-level tool discussion in parallel) • 1400–2000 HLS 6.2.5 … 6.2.8 (Rm. 202, chaired by Jill Boyce) Sat. 28 June, 3rd day • 0900–1330 Viewing sessions for CfE dry-run • 0900–1300 HLS 6.2.2, 6.2.10, revisits, 6.3, 6.4 (Rm. 202, chaired by Jill Boyce) • 1430–1900 Viewing sessions for CfE dry-run • 1400–2000 HLS 6.2.5, revisits, 6.3, 6.4 (Rm. 202, chaired by Jill Boyce) • 1500–1730 ITU-T Q6/21 VCEG (outside of JVET, Rm. 201) Sun. 29 June, 4th day • 0800–1230 EE1 related (5.1.3/5.1.4) (Rm. 201) • 0900–1300 HLS 6.x (Rm. 202, chaired by Jill Boyce) • 1400–2000 Non-EE2 (5.2.4 (Rm. 201, chaired by Yan Ye from 1750 onwards) • 1400–2000 HLS 6.x (Rm. 202, chaired by Jill Boyce) Mon. 30 June, 5th day • 0900–1200 MPEG information sharing session • 1300–1350 JVET plenary: Coordination, planning, reports/issues from tracks (Rm. 201) • 1400–1700 Review CfE related topics from AHG7 (4.8) and AHG17 (4.16) (Rm. 201) • 1700–1800 Joint with AG 5: Review CfE dry-run results, Multi-layer verification testing (Rm. 201) • 1400–1800 HLS 6.x (Rm. 202, chaired by Jill Boyce) • 1800–2030 Review topics from CICP (4.18), AHG18 (4.17) Tue 1 July, 6th day • 0900–1300 Continue review CfE related topics (4.16), EE1/2 remaining docs and revisits (Rm. 201) • HLS remaining topics & revisits (Rm. 202, chaired by Jill Boyce) • 1400–1700 JVET main session: Remaining 4.17 document review (Rm. 201) • 1400–XXXX BoG on tool complexity analysis (Rm. 202, chaired by Xiang Li) • XXXX–XXXX BoG on CfE updates (Rm. 202, chaired by Frank Bossen) Wed. 2 July, 7th day • 0900–1015 MPEG information sharing session • 1030–1300 JVET main session: Coordination, planning, BoG reports, report/revisits from HLS, CICP • 1400–1600 JVET main session: CICP cont., revisits 4.16, CfE review & planning, • 1615-1645 Joint with WG 4 and WG 7 on Gaussian splat coding Thu. 3 July, 8th day • 0900–1200 JVET main session: CfE review & planning, • 1200–1330 JVET main session: Doc review 4.3 and 4.4 • 1400–1600 Joint with VCEG, WG 2, AG 5: next generation video standardization (Rm. 201) • 1610–1900 JVET plenary: DoCR and other output doc review/planning, AHG planning, remaining doc review 4.x, revisits (chaired by Y. Ye from 1815)
2024.07.01 (화)	- 박병주 석사과정 • 대전역 KTX → 부산역 KTX: 15:05 ~ 16:35 (1 시간 30 분)
2024.07.03 (목)	- 김선재/이태식 박사과정, 이동호/정현아 석사과정

	• 대전역 KTX → 부산역 KTX: 16:39 ~ 18:03 (1 시간 24 분)

2. 주요 활동 내용 (JVET)

2.1. AHG11 Neural Network-based Video Coding (NNVC)

2.1.1. 회의 내용 및 결과

❍ **AHG report 검토 [JVET-AM0011] [Co-chairs: E. Alshina (Huawei), F. Galpin (InterDigital), S. Liu (Tencent), A. Segall (Amazon), Vice-chairs: J. Li (ByteDance), R.-L. Liao (Alibaba), M. Santamaria (Nokia), T. Shao (Dolby), M. Wien (RWTH), P. Wu (ZTE)]**

- Evaluate and quantify the performance improvement potential of NN-based video coding technologies compared to existing video coding standards such as VVC, including both individual coding tools, architectures and content adaptation with NN parameters overfitting.
- Establish logistics of training data sets, including list, location and md5sums. Generate and distribute anchor encoding, and develop supporting software as needed.
- Study potential improvements of the NNVC CTC document JVET-AJ2016.
- Study the impact of training (including the impact of loss functions) on the performance of candidate technologies and identify suitable material for testing and training.
- Analyze complexity characteristics for technologies under study, including transformers, perform complexity analysis, and develop complexity reductions of candidate technology. Discuss and propose improved metrics to perform complexity analysis of NN architectures, in particular investigate bit-exact reproducibility of NN-based methods on various platforms.
- Finalize and discuss the EE on neural network-based video coding.
- Promote the call for training materials, distribute it, and actively communicate with content owners.
- Coordinate with other groups, including SC 29/AG 5 on the evaluation and assessment of visual quality, and AHG12 on the interaction with ECM coding tools. If possible, prepare encodings with combinations of tools included in the NNVC software for visual quality assessment at the next meeting.
- Coordinate with AHG14 on items related to NNVC software development and implement to NNVC software branch and design evaluation tests for end-to-end AI

coded reference picture.

❍ **NNVC Software Development AHG14 report 검토 [JVET-AM0014] [F. Galpin (InterDigitla), R. Chang (Tencent), Y. Li (ByteDance), Y. Li (Qualcomm), M. Santamaria (Nokia), J. N. Shingala (Ittiam), Z. Xie (OPPO)]**

- Software development
  - Git**:** https://vcgit.hhi.fraunhofer.de/jvet-ahg-nnvc/VVCSoftware_VTM
  - VTM-11.0 기반 소프트웨어에 아래의 NNVC 기능 추가
    A. MCTF (Motion Compensated Temporal Filtering) 활성화: JVET-V0056
    B. GOP32
    C. RDO 내 Deblocking 기능 활성화
    D. Improved MCTF: JVET-AH0054
    E. Reference Picture Alignment 개선: JVET-AI0124
  - Software changes
    A. Several commits were merged in the NNVC repository. The following changes were integrated:

Contribution	MR	Description
AL0084	MR 303	LOP6
AL0169	MR 304	ALOP
	MR 302	Various cleaning and fixes
AL0265	MR 306	Sadl v13

B. In SADL v13, the following changes were integrated:

Contribution	MR	Description
	MR 152	Optimization for multi-output models
	MR 151	Extend broadcasting for add layer
AL0100	MR 150	Tile feature
AL0100	MR 149	SIMD optimization for 2x2 deconvolution

- Comparison to VTM
  - NNVC-12.0 VTM vs NNVC-13.0 VTM
    A. VTM configuration of NNVC is not changed between version 10.0 and 13.0.
  - NNVC-13.0 VTM vs NNVC-13.0 anchor (Overall, CPU)

BD-rate Over NNVC-13.0 VTM									
All Intra Main10					Random access Main10				
Y	U	V	EncT	DecT	Y	U	V	EncT	DecT
-8.21%	-14.90%	-13.48%	113%	2813%	-9.19%	-15.68%	-15.49%	157%	2296%
Low delay B Main10					Low delay P Main10				
Y	U	V	EncT	DecT	Y	U	V	EncT	DecT
-6.26%	-9.59%	-9.71%	106%	3028%	-6.80%	-11.98%	-11.76%	110%	3075%

- NNVC-13.0 VTM mode vs NNVC-13.0 HOP.5
  A. The NNVC-13.0 did not change the HOP5 model. Performance is the same as NNVC-11.0 with a slight decoding time increase.
- NNVC-13.0 VTM mode vs NNVC-13.0 VLOP
  A. The NNVC-13.0 did not change the VLOP3 model. Performance is the same as NNVC-11.0 with a slight decoding time decrease.
- NNVC-13.0 VTM vs NNVC-13.0 NNSR
  A. The NNVC-13.0 did not change the NNSR model. Performance is the same as NNVC-11.

• Comparison to NNVC-13.0 anchor

- NNVC-12 anchor vs NNVC-13.0 anchor (Overall, CPU)

BD-rate Over NNVC-12.0 VTM									
All Intra Main10					Random access Main10				
Y	U	V	EncT	DecT	Y	U	V	EncT	DecT
0.02%	0.15%	-0.02%	99%	97%	0.01%	0.38%	0.02%	99%	97%
Low delay B Main10					Low delay P Main10				
Y	U	V	EncT	DecT	Y	U	V	EncT	DecT
-0.07%	1.66%	-0.39%	98%	99%	0.06%	1.33%	-0.49%	94%	94%

• Performance of new NNVC development branch

- The new version benefits from VTM improvement between version 11.0 and 23.9.
- NNVC-13.0 VTM vs NNVC-new VTM

BD-rate Over NNVC-13.0 VTM									
All Intra Main10					Random access Main10				
Y	U	V	EncT	DecT	Y	U	V	EncT	DecT
-0.55%	-0.43%	-0.17%	85%	96%	-0.10%	0.18%	0.21%	71%	92%
Low delay B Main10					Low delay P Main10				
Y	U	V	EncT	DecT	Y	U	V	EncT	DecT
-0.11%	0.23%	0.19%	72%	90%	N/A	N/A	N/A	N/A	N/A

- NNVC-13.0 VTM vs NNVC-new VTM

BD-rate Over NNVC-6.0 VTM									
All Intra Main10					Random access Main10				
Y	U	V	EncT	DecT	Y	U	V	EncT	DecT
0.00%	0.00%	0.00%	105%	119%	0.00%	0.00%	0.00%	101%	146%

- NNVC-13.0 vs NNVC-new

BD-rate Over NNVC-13.0 VTM	
All Intra Main10	Random access Main10

Y	U	V	EncT	DecT	Y	U	V	EncT	DecT
0.06%	0.03%	0.11%	86%	98%	0.07%	0.18%	0.12%	73%	97%
Low delay B Main10					Low delay P Main10				
Y	U	V	EncT	DecT	Y	U	V	EncT	DecT
0.00%	0.04%	0.29%	74%	98%	N/A	N/A	N/A	N/A	N/A

- NNVC-new vs NNVC-new-VLOP, HOP

BD-rate Over NNVC-new									
Random access Main10									
VLOP					HOP				
Y	U	V	EncT	DecT	Y	U	V	EncT	DecT
2.63%	10.14%	9.16%	98%	47%	-6.60%	-5.97%	-7.57%	290%	4253%

❍ **EE (Exploration experiment) Activity 검토 [JVET-AM0023] [E. Alshina (Hauwei), R. Chang (Tencent), F. Galpin (InterDigital), Y. Li (ByteDance), Y. Li (Qualcomm), M. Santamaria (Nokia), T. Shao (Dolby), J. Ström (Ericsson), Z. Xie (OPPO) (EE coordinators)]**

- 직전 회의 (JVET-AL)에서 아래와 같은 EE를 진행하기로 결정하였으며, 금번 회의에서 해당 EE 결과들을 보고함
  - EE1-1: LOP in-loop filter
    1. EE1-1.1 – Sample-based adaptive blending weight selection for LOP JVET-AM0123
    2. EE1-1.2 – Over-Parameterized LOP In-Loop Filter JVET-AM0120
    3. EE1-1.3 – Backbone Block Enhancement of LOP In-Loop Filter with Over-Parameterized Training and Variable Channels JVET-AM0122
    4. EE1-1.4 – Conditional loop-filter (withdrawn)
  - VLOP in-loop filter
    1. EE1-2.1 – Improved VLOP with Attention JVET-AM0135
    2. EE1-2.2 – Conditional loop-filter JVET-AM0053
  - EE1-3: NN-inter prediction
    1. EE1-3.1 – RA/LDB Unified Reference Frame Synthesis for VVC Inter Coding JVET-AM0114
    2. EE1-3.2 – Deep Reference Frame Generation for Inter Prediction Enhancement JVET-AM0175
  - EE1-4: NN-based super-resolution
    1. EE1-4.1 – Cross-component enhanced NNSR JVET-AM0257
- 각 EE들에 대한 결과 요약은 아래 표와 같음 (Anchor: NNVC VTM mode)
  - All Intra cfg BD-rate (vs VTM)

Test	All Intra cfg BD-rate vs VTM	kMAC/pxl	Total Num Param

	Y	U	V	Enc	Dec		
NN-Intra & LOP6 NN-filter							
NNVC-LOP6	0.0%	0.0%	0.0%	100%	100%	21.4	1.5
EE1-1.1.2	0.0%	-0.2%	-0.2%			21.4	1.5
EE1-1.1.3	-0.1%	-0.6%	-0.2%			21.4	1.5
* Tests EE1-1.2 and 1.3 require change of training. Due to the over-parametrization training time increases from 170 to 180 (EE1-1.2) and 190 (EE1-1.3) hours. Run time is not reported.							
NN-Intra & VLOP3 NN-filter							
NN-Intra & NN-Inter							
NN-Intra & LOP6 NN-filter & NN-super resolution							
NNVC-LOP6	-8.4%	-14.2%	-13.9%	100%	100%	26.1	21.1
EE1-4.1	-8.6%	-11.8%	-11.3%	105%	N/A	26.6	29.0
* Dec Run time is not reported, only all-intra test conducted. BD-rate is very content dependent, varies in a range -3%...1%.							

➢ Low-delay cfg BD-rate (vs VTM)

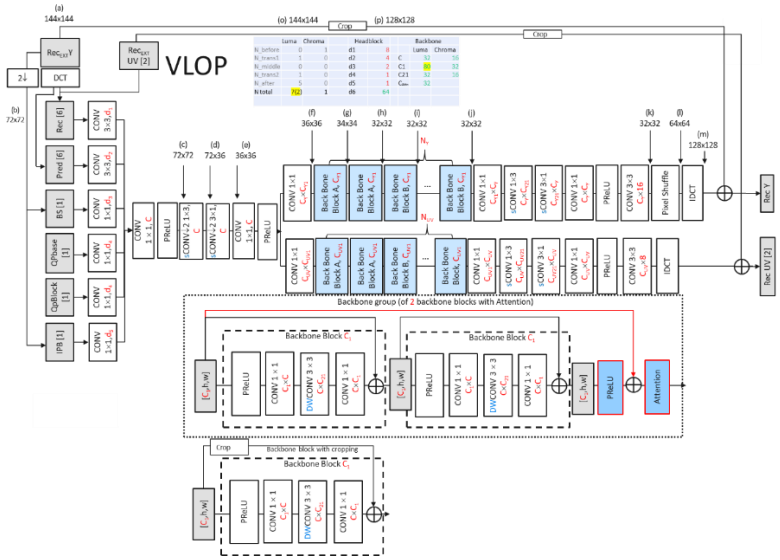
Test	Low-delay cfg BD-rate vs VTM					kMAC/pxl	Total Num Param
	Y	U	V	Enc	Dec		
NN-Intra & LOP6 NN-filter							
NNVC-LOP6	0.0%	0.0%	0.0%	100%	100%	21.4	1.5
EE1-1.1.1	-0.1%	-0.7%	-0.8%	102%	101%	21.4	1.5
EE1-1.1.2	-0.1%	-0.8%	-0.9%	100%	102%	21.4	1.5
EE1-1.1.3	-0.1%	-1.0%	-1.3%	101%	102%	21.4	1.5
* Tests EE1-1.1 do not require training. Only filter use has been changed, by modifying blending weights at sample level.							
NN-Intra & VLOP3 NN-filter							
NNVC-VLOP3	0.0%	0.0%	0.0%	100%	100%	9.9	1.4
EE1-1.2.1	-0.1%	-0.5%	-1.0%	100%	102%	9.9	1.4
EE1-1.2.2	-0.1%	-0.3%	-1.2%	100%	102%	9.9	1.4
EE1-2.1	-0.3%	-1.9%	0.0%	102%	113%	9.9	1.4
EE1-2.2.1	0.0%	-0.8%	-2.0%	100%	109%	10.7	1.4
EE1-2.2.2	0.1%	-1.1%	-2.1%	101%	103%	9.9	1.4
* Tests EE1-1.1 and EE1-2.2 do not require training. Only filter use has been modified. N the worst case some operations (head and fusion blocks) in EE1-2.2.1 design will be executed twice, additional complexity is 0.8 kMAC/pxl so total complexity of filter is 5.9 kMAC/pxl. * In test EE1-2.1 attention module (similar to one in LOP) has been added to VLOP. Training time reported to be 240 hours. In variant of EE1-2.1 with improved SIMD code, decoding run time increase is lower (not 110%, but 103% in RA test).							
NN-Intra & NN-Inter							
NNVC-LOP6	0.0%	0.0%	0.0%	100%	100%	21.4	1.5
EE1-3.1	-3.3%	-4.9%	-4.1%	102%	N/A	508	7
EE1-3.2						525	5.3
* No decoding run time reported for EE1-3.1. Increases of computational complexity is significant. * Code for EE1-3.2 made available after deadline, no LD-B results provided.							
NN-Intra & LOP6 NN-filter & NN-super resolution							

➢ Random Access cfg BD-rate (vs VTM)

Test	Random Access cfg BD-rate vs VTM					kMAC/pxl	Total Num Param
	Y	U	V	Enc	Dec		
NN-Intra & LOP6 NN-filter							
NNVC-LOP6	0.0%	0.0%	0.0%	100%	100%	21.4	1.5
EE1-1.1.1	-0.1%	-0.2%	-0.2%	100%	103%	21.4	1.5
EE1-1.1.2	-0.1%	-0.2%	-0.2%	99%	103%	21.4	1.5
EE1-1.1.3	-0.1%	-0.2%	-0.2%	99%	103%	21.4	1.5
* Tests EE1-1.1 do not require training. Only filter use has been changed, by modifying blending weights at sample level.							
NN-Intra & LOP6 NN-filter							
NNVC-LOP6	0.0%	0.0%	0.0%	100%	100%	21.4	1.5
EE1-1.2	0.0%	0.1%	-0.2%			21.4	1.5
EE1-1.3	-0.1%	-0.5%	0.1%	100%	92%	21.4	1.5
* Tests EE1-1.2 and 1.3 require change of training. Due to the over-parametrization training time increases from 170 to 180 (EE1-1.2) and 190 (EE1-1.3) hours. Run time is not reported.							
NN-Intra & VLOP3 NN-filter							
NNVC-VLOP3	0.0%	0.0%	0.0%	100%	100%	9.9	1.4
EE1-1.2.1	0.0%	0.0%	0.0%	99%	101%	9.9	1.4
EE1-1.2.2	0.0%	0.1%	0.1%	100%	101%	9.9	1.4
EE1-2.1	-0.5%	-1.1%	-0.4%	99%	110%	9.9	1.4
EE1-2.2.1	0.0%	-0.2%	-0.1%	100%	113%	10.7	1.4
EE1-2.2.2	0.1%	0.0%	0.0%	101%	100%	9.9	1.4
* Tests EE1-1.1 and EE1-2.2 do not require training. Only filter use has been modified. N the worst case some operations (head and fusion blocks) in EE1-2.2.1 design will be executed twice, additional complexity is 0.8 kMAC/pxl so total complexity of filter is 5.9 kMAC/pxl. * In test EE1-2.1 attention module (similar to one in LOP) has been added to VLOP. Training time reported to be 240 hours. In variant of EE1-2.1 with improved SIMD code, decoding run time increase is lower (not 110%, but 103% in RA test).							
NN-Intra & NN-Inter							
NNVC-LOP6	0.0%	0.0%	0.0%	100%	100%	21.4	1.5
EE1-3.1	-4.2%	-4.8%	-5.4%	101%	N/A	508	7
EE1-3.2	-1.9%	-1.0%	-1.0%	133%	2188%	525	5.3
* No decoding run time reported for EE1-3.1. Increases of computational complexity is significant. * Code for EE1-3.2 made available after deadline, no LD-B results provided.							
NN-Intra & LOP6 NN-filter & NN-super resolution							
NNVC-LOP6	0.0%	0.0%	0.0%	100%	100%	26.1	21.1
EE1-4.1						26.6	29.0
* Dec Run time is not reported, only all-intra test conducted. BD-rate is very content dependent, varies in a range -3%...1%.							

- 금번 회의에서 발표한 루프내 필터 관련 NNVC EE의 주요결과와 기술요약은 아래와 같음
  - VLOP3 EE1-2.1 Adopt: Improved VLOP Attention with SIMD acceleration
  - NN-Inter EE1-3.2 Adopt: Deep Reference Frame Generation for Inter Prediction Enhancement

EE1	Title	Contributors	Reviews

2.1 (VLOP)	Improved VLOP Attention with SIMD acceleration (AM0135)	Y. Li M. Coban M. Karczewicz L. Kerofsky (Qualcomm)	**Summary** • 기존 VLOP3 구조를 기반으로 attention module 개선 및 SIMD acceleration을 적용한 VLOP 필터(Int16 모델) 기고 • Luma branch에는 attention 포함 residual backbone 구조 적용, chroma branch는 backbone block만 사용 • Attention은 max pooling → depth-wise convolution → bilinear upsampling → hard sigmoid → element-wise 곱 연산으로 구성 • Backbone block은 2개의 cropping 포함 블록 후 [2,3] 구조의 그룹별 block으로 구성 • 전체 모델 복잡도는 5.09 kMAC/pixel 수준이며, 54개의 convolution 계층으로 구성됨 • 추론은 SADL 프레임워크(Int16 모델) 기반, 학습은 PyTorch 기반(VLOP3 training strategy 사용) • 학습 데이터: DIV2K, BVI, TVD / Loss: L1, L2 / QP: {22,27,32,37,42} / GPU: Tesla A100 • SADL 연산자에 대해 SIMD 최적화 구현 포함 (자세한 사항은 JVET-AM0136 참고) **Experimental results {Y, U, V, EncT, DecT}** ➢ **vs VLOP3 anchor (NNIntra ON)** · AI: -0.5%, -0.7%, -0.7%, 100%, 105% · RA: -0.5%, -1.1%, -0.4%, 100%, 103% ➢ **vs VTM** · AI: -7.8%, -9.4%, -9.0% · RA: -6.2%, -7.6%, -6.1%  Figure 1 The proposed VLOP with attention architecture.

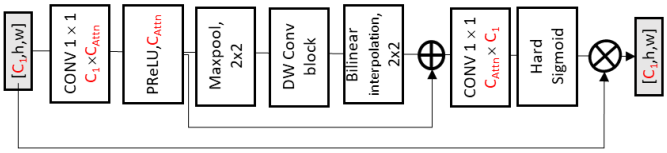


Figure 2 The Attention module.

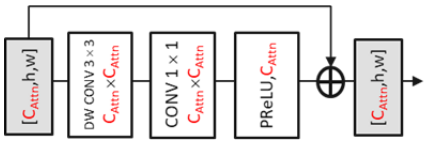


Figure 3 The depth-wise convolution block (DW Conv block) for the Attention module.

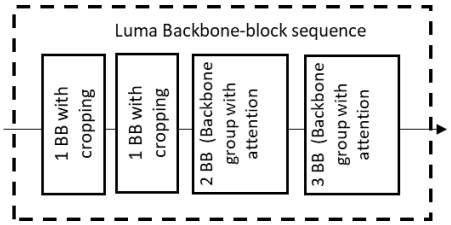


Figure 4 The configuration overview of cropping BB and attention BB groups for luma branch.

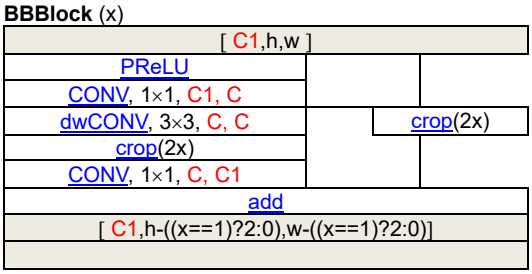


Figure 5. Diagram in detail for the backbone block.

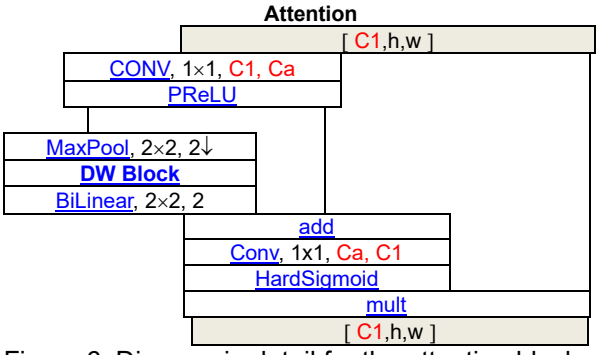
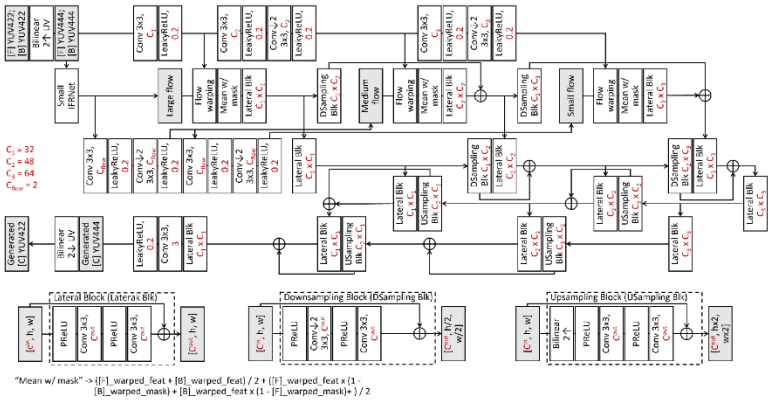


Figure 6. Diagram in detail for the attention block.

Comments

• 기존 LOP에서 사용된 attention 메커니즘을 VLOP에 적용했으나, 이전 실험에서 Loss가 발생하였음

• 초기에는 coding gain이 약 0.5% 향상되었으나, DecT. 이 증가해 우려 제기됨

			• 이후 7월 1일 이후 논의(v2 결과)에서 SIMD 최적화를 시도함 → CPU EncT. +1%, DecT. +7% gain이 발생함(GPU에서는 anchor보다 빠름) • 이를 바탕으로 SIMD 최적화가 적용된 attention 구조는 수용 가능한 Trade-off를 제공함 • 유사한 attention 구조로 인해 LOP에서도 DecT. 증가가 관찰되었으며, 이에 따라 LOP에도 SIMD 최적화 적용 가능성 제안 **Decision** **• Adopt JVET-AM0135 EE1-2.1**
3.2 (NN-Inter)	Deep Reference Frame Generation for Inter Prediction Enhancement (AM0175)	Xinxin Chen, Nianxiang Fu, Wenzhuo Zhang, Junxi Zhang, Ding Ding, Wenzhuo Ma, Zhenzhong Chen (Wuhan University)	**Summary** • 기존 JVET-AL0184 기반의 deep reference frame(DRF) 생성 구조를 개선하여 int16 SADL 기반 추론 방식으로 구현함 • 생성된 reference frame은 기존 프레임을 대체하지 않고 추가로 삽입하는 방식(JVET-AF0208)을 채택하여 inter prediction 성능 개선 • 인코딩 시 2장의 복원 프레임을 입력으로 사용해 새로운 참조 프레임을 생성, 해당 프레임을 RPL에 삽입하여 현재 프레임 예측에 활용함 **Experimental results {Y, U, V, EncT, DecT}** ➤ NNVC-13.0 기준, int16 SADL 모델 (제안 방식 vs anchor) – AI config 미제공, RA 기준 전체 평균 BD-rate: · Y: -1.85%, U: -0.99%, V: -0.97%, EncT: 133%, DecT: 2188% ➤ 비교 실험: – float32 vs int16 모델: BD-rate는 감소했지만 EncT, DecT 감소  Fig. 1: The architecture of the DRF network **Comments**

			• 초기 구현은 fully integerized로 보고되었으며, Vimeo90K 기반 학습 및 학습 스크립트까지 포함되어 cross-check 완료됨 • 다만, runtiome trade-off는 좋지 않았음 → NNVC에 없던 새로운 요소이므로 초기 연구 기반으로 유용하다고 판단 • 이후 구현이 완전한 정수형이 아니었던 점이 보고되었고, software coordinator의 지원으로 해당 문제가 수정됨 • 수정 이후에는 작지만 성능 향상도 확인되어 긍정적으로 평가됨 **Decision** **• Adopt JVET-AM0175 EE1-3.2 (non CTC, also provide training script)**

❍ **Next EE [JVET-AM2023] [E. Alshina, R. Chang, F. Galpin, Y. Li, Y. Li, M. Santamaria, T. Shao J. Ström, and Z. Xie (EE coordinators)]**

- 금번 회의에서 진행한 EE activity review 및 non-EE 기고 review를 통해 차기 회의 (JVET-AN)에서 진행할 EE를 결정함
- LOP 3건, VLOP 2건, NN-inter 4건, NNSR 2건의 신규 EE 및 Next EE가 할당됨
- 차기 회의 EE를 요약한 테이블은 아래와 같음

EE1	Tests	Tester
LOP		
1.1	LOP with Overlapped Feature Integration (JVET-AM0131)	UESTC Transsion
1.2	Backbone Block Enhancement of LOP In-Loop Filter with Over-Parameterized Training and Variable Channels (JVET-AM0122)	Xidian Univ.
1.3	Conditional LOP loop-filter (JVET-AM0053)	Nokia
VLOP		
2.1	Conditional VLOP loop-filter (JVET-AM0053)	Nokia
2.2	Decomposed Content-Adaptive VLOP (JVET-AM0185)	TCL
NN-Inter		
3.1	Deep Reference Frame Generation for Inter Prediction Enhancement with Structural Re-parameterization (JVET-AM0177)	Wuhan Univ.
3.2	RA/LDB Reference Frame Synthesis for VVC Inter Coding (JVET-AM0115)	Xidian Univ.
3.3	Network architecture from EE1-3.1 with training from EE1-3.2	N/A
3.4	Network architecture from EE1-3.2 with training from EE1-3.3	N/A
NN-based super-resolution		
4.1	NNSR with new backbone block based on Spatial-Channel Mixing (SCM) (JVET-AM0199)	Kyung Hee Univ. ETRI
4.2	Cross-component enhanced NNSR (JVET-AM0257)	HUST Vivo

2.2. AHG12 Enhanced Compression Beyond VVC Capability

2.2.1. 회의 내용 및 결과

❍ **AHG report 검토 (M. Karczewicz, Y. Ye, L. Zhang, B. Bross, X. Li, R. Chernyak, K. Naser, Y. Yu)**

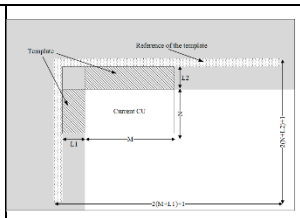
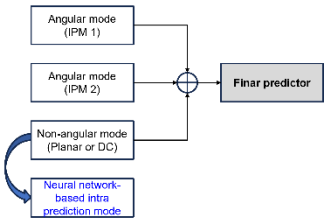
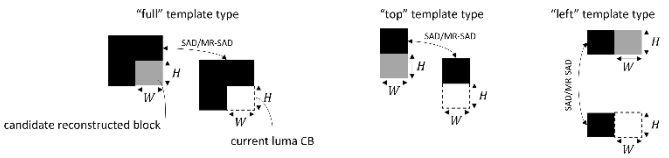
- Mandates
  - Solicit and study non-neural-network video coding tools with enhanced compression capabilities beyond VVC.
  - Discuss and propose refinements to the ECM17 algorithm description JVET-AL2025.
  - Coordinate with AHG7 to study the performance and complexity tradeoff of these video coding tools.
  - Coordinate with AHG6 on ECM software development.
  - Support AHG6 in generating anchors according to the test conditions in JVET-AI2017.
  - Analyse the results of exploration experiments described in JVET-AL2024 in coordination with the EE coordinators.
  - Coordinate with AHG11 to study the interaction with neural network-based coding tools.

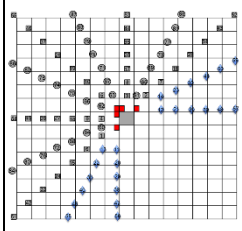
❍ **EE (Exploration experiment) Activity 검토**

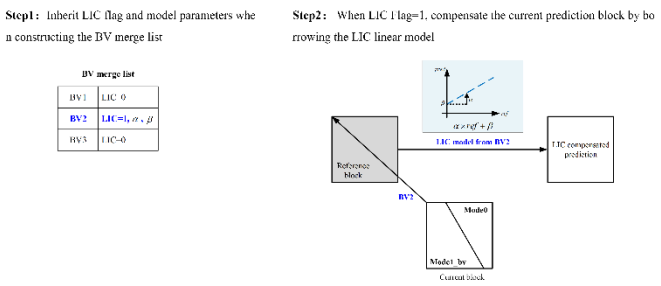
- 지난 38차 JVET 회의에서 제안된 부호화 툴에 대한 검토 후 4개의 EE를 구성하였으며, 이번 39차 JVET 회의에 제출된 EE 실험결과 (압축성능, 부/복호화기 복잡도, 기존 Tools간 interaction)을 검토하여 ECM-18.0 SW 채택 여부를 결정하였음.

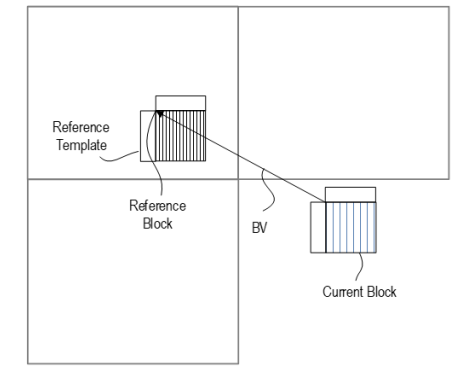
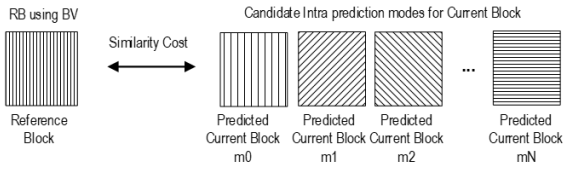
- **Intra Prediction (10건)**

EE2	Title	Coordinators	Reviews
1.1	TIMD fusion with neural network based intra prediction	Y.-H. Lin (Sharp) K. Naser (InterDigital)	**Summary** • TIMD를 수행하는 과정에서 도출된 IPM과 non-angular model를 blending 할 때, non-angular mode를 neural network based intra prediction로 사용하는 방법이 기고됨

			  • 비슷한 기고인 EE2-1.2와 비교했을 때 성능적인 측면에서 이득이 없음 • Experimental results {Y, U, V, EncT, DecT} ➢AI: -0.01%(Y), 0.01%(U), 0.03%(V), 100.3%(EncT), 100.9%(DecT) ➢RA: -0.02%(Y), -0.10%(U), 0.11%(V), 100.2%(EncT), 100.4%(DecT) **Decision** **• No action**
1.2	TIMD fusion with block vector based prediction	J. Fu (PKU) K. Naser (InterDigital) D. Ruiz Coll (Ofinno)	**Summary** • TIMD를 수행하는 과정에서 도출된 IPM과 non-angular model를 blending 할 때, non-angular mode를 BV기반 블록으로 사용하는 방법이 기고됨 • Test 1.2a: Non angular mode 를 BV기반 블록으로 대체 • Test 1.2b: BV merge list construction enhancement • Test 1.2c: Test 1.2a + Test 1.2b • Experimental results {Y, U, V, EncT, DecT} # EE2-1.2a: ➢AI: {-0.01%, -0.05%, -0.01%, 99.7%, 100.2%, 101.0%, 100.2%} ➢RA: {-0.01%, 0.01%, 0.01%, 100.0%, 100.8%, 100.0%, 100.2%} # EE2-1.2b: ➢AI: {-0.05%, -0.05%, -0.03%, 99.7%, 101.7%, 100.9%, 100.5%} ➢RA: {-0.01%, 0.05%, 0.03%, 99.6%, 100.7%, 100.2%, 100.4%} # EE2-1.2c: ➢AI: {-0.07%, -0.06%, -0.05%, 99.9%, 101.0%, 101.1%, 100.7%} ➢RA: {-0.03%, 0.00%, 0.01%, 100.3%, 100.1%, xx%, xx%} **Decision** **• Adopt JVET-AM0307 test 1.12.**
1.3	IntraTMP template type adaptation	T. Dumas (InterDigital)	**Summary** • IntraTMP 수행 시 사용되는 template type을 list에 함께 저장하는 방법이 기고됨  Index / BV / SAD 0 / BV_x, BV_y / SAD_0 1 / BV_x, BV_y / SAD_1 2 / BV_x, BV_y / SAD_2 3 / BV_x, BV_y / SAD_3 … / … / … 19 / BV_x, BV_y / SAD_{29} Index / BV / Template / SAD 0 / BV_x, BV_y / full / SAD_0 1 / BV_x, BV_y / top / SAD_1 2 / BV_x, BV_y / top / SAD_2 3 / BV_x, BV_y / full / SAD_3 … / … / … / … 19 / BV_x, BV_y / left / SAD_{29}

			• Experimental results {Y, U, V, EncT, DecT} ➢AI: -0.04% -0.05% -0.01% 99.9% 100.0% ➢ RA: -0.01% -0.15% -0.01% 100.1% 100.0% **Decision** **• Adopt JVET-AM0307 test 1.12**
1.4	Test 1.2c + Test 1.3	J. Fu (PKU) K. Naser (InterDigital) D. Ruiz Coll (Ofinno)	**Summary** • Test 1.2c와 Test 1.3의 combination test 가 기고됨 • Experimental results {Y, U, V, EncT, DecT} ➢AI: -0.12% -0.08% -0.08% | 100.3% EncT and 102.2% DecT ➢RA: -0.04% -0.01% 0.02% | 100.2% EncT and 100.4% DecT. **Decision** **• No action**
1.5	Test 1.1 + Test 1.4	J. Fu (PKU) K. Naser (InterDigital) D. Ruiz Coll (Ofinno) Y.-H. Lin (Sharp)	**Summary** • Test 1.1과 Test 1.4의 combination test 가 기고됨 • Experimental results {Y, U, V, EncT, DecT} ➢AI: -0.13% -0.14% -0.11% | 100.6% EncT and 103.4% DecT ➢RA: -0.06% 0.00% 0.01% | 100.3% EncT and 100.5% DecT. **Decision** **• No action**
1.6	Intra merge mode	M. Blestel (Ofinno) Y.-J. Chang(Qualcomm) J. Fu (PKU)	**Summary** • 인접 블록으로부터 Intra Prediction Mode (DIMD, OBIC, TIMD, SGPM, TMRL, IntraTMP)를 취합해 merge list 생성하는 방법이 기고됨 • 각 merge index는 최대 8개 IPM, weights, location dependent weight, BV, transform type을 가질 수 있음 • Merge list 크기는 5로 지정, DIMD sub-mode로 signaling 수행 • Test 1.6a: Intra merge mode with inter non-adjacent pattern • Test 1.6b: Intra merge mode with OBIC non-adjacent pattern  **Spatial adjacent map** **Intra merge list** | Index | Intra Prediction Mode | | --- | --- | | 0 | **DIMD** (IPM1, IPM2, … IPM8) | | 1 | **DIMD** (IPM1, IPM2, … IPM8) | | 2 | **OBIC** (IPM1, IPM2, … IPM5) | | 3 | **TIMD** (IPM1, IPM2, IPM3) | | 4 | **SGPM** (IPM1, IPM2) | • Experimental results {Y, U, V, EncT, DecT} Test 1.6A ➢AI: {-0.03%, 0.2%, 0.3%, 100.1%, 99.6%}. ➢RA: {-0.01%, 0.09%, 0.09%, 99.2%, 100.1%}. Test 1.6B ➢AI: {-0.03%, -0.09%, -0.026%, 100.0%, 99.7%}. ➢RA: {0.01%, -0.03%, 0.15%, 99.1%, 100.1%}. **Decision** **• Adopt JVET-AM0307 test 1.12**

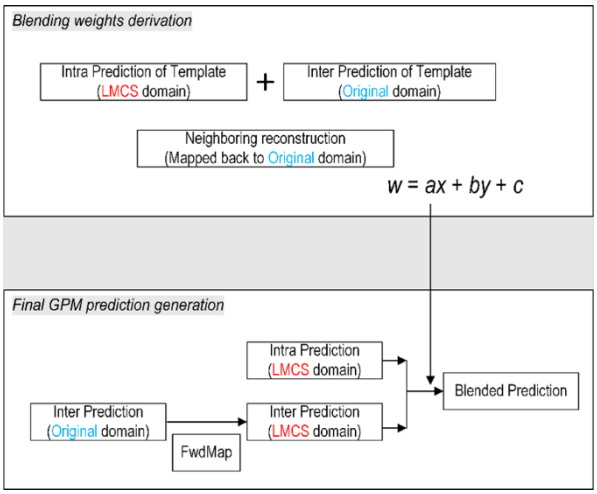
1.7	Block prediction with cubic interpolation filter in TIMD	H. Qin (TCL)	**Summary** • TIMD의 template 영역 predictor 생성 및 현재 블록 영역 predictor 생성에 사용하는 interpolation filter를 변경하는 방법이 기고됨 기존 방법 (ECM-16.1) • Template 영역: 4-tap cubic filter • 현재 블록 영역: 8-tap cubic filter or 8-tap gaussian filter Test 1.7a • Template 영역: 4-tap cubic filter • 현재 블록 영역: 8-tap cubic filter Test 1.7b • Template 영역: 6-tap cubic filter • 현재 블록 영역: 8-tap cubic filter or 8-tap gaussian filter • Experimental results {Y, U, V, EncT, DecT} Test 1.7a: ➢AI: {-0.02%, -0.05%, -0.01%; 100.0%, 99.7%}; ➢RA: {-0.01%, 0.09%, 0.08%; 100.0%, 100.1%}. Test 1.7b: ➢AI: {0.01%, 0.03%, 0.02%; 99.8%, 99.8%}; ➢RA: {0.00%, 0.06%, 0.10%; 100.1%, 100.1%}. **Decision** **• Adopt JVET-AM0163 test 1.7a**
1.8	Harmonization of SGPM-BV and LIC	J. Huo, Y. Fei (Xidian Univ)	**Summary** • SGPM 수행 시 LIC flag와 model parameters를 inherit 하는 방법이 기고됨  • Screen contents에서 성능에 비해 Nature contents에서는 gain이 없음 • Experimental results {Y, U, V, EncT, DecT} For AI configuration: ➢Natural: 0.00%, 0.01%, 0.02%, with 100.6% EncT, 100.5% DecT. ➢Class F: -0.21 %, -0.20 %, -0.36 %, with 100.1% EncT, 100.4% DecT. ➢Class TGM: -0.04 %, -0.06 %, -0.07 %, with 100.0% EncT, 100.1% DecT. For RA configuration: ➢Class F: -0.09 %, -0.21 %, -0.28 %, with 99.8% EncT, 100.1% DecT. ➢Class TGM: -0.06%, -0.04 %, -0.07 %, with 99.9% % EncT, 100.0% DecT. **Decision** **• Adopt JVET-AM0157 test 1.8**

1.9	Block vector-based intra mode derivation	J.-K. Lee (Ofinno)	**Summary** • BV로 찾은 참조블록과 현재 블록의 MPM을 비교하여 가장 작은 SAD를 가지는 2개의 IPM을 유도하는 방법이 기고됨   • Luma에서 작은 gain이 있으나 chroma에서 loss가 있음 • Experimental results {Y, U, V, EncT, DecT} ➢AI: Overall {-0.02%, 0.14%, 0.26%, 100.9%(EncT), 99.4%(DecT)}. ➢RA: Overall {0.00%*, -0.03%, 0.13%, 97.3%(EncT), 101.0%(DecT)} **Decision** • **No action**
1.10a	Matrix-based position dependent intra prediction for GPM/CIIP	Z. Sun (OPPO)	**Summary** • 현재 블록이 MPDIP에 적합한 경우, (디코딩된 왼쪽과 위의 참조 샘플이 사용 가능하고 현재 블록 크기가 MPDIP를 지원하는 경우) GPM-IPM 목록의 일부 non-MPDIP 모드는 인접한 MPDIP 모드로 대체되는 방법이 기고됨 • Test 1.10a: Matrix-based position dependent intra prediction for GPM. • Test 1.10b: Matrix-based position dependent intra prediction for CIIP. • Test 1.10c: Test 1.10a + Test 1.10b. • Test 1.10d: Extension to regression GPM. • Experimental results {Y, U, V, EncT, DecT} EE2-1.10-a: ➢RA {-0.01% Y, 0.07% U, 0.02% V, 100.0% EncT, 100.1% DecT } ➢LB {-0.01% Y, 0.22% U, 0.08% V, 99.3% EncT, 99.3% DecT} EE2-1.10-b: ➢RA {-x.xx% Y, -x.xx% U, -x.xx% V, xx.x% EncT, xx.x% DecT } ➢LB {0.03% Y, 0.35% U, 0.31% V, 99.8% EncT, 100.0% DecT} EE2-1.10-c: ➢RA {0.00% Y, 0.07% U, 0.02% V, 99.8% EncT, 100.4% DecT } ➢LB {0.00% Y, 0.31% U, -0.07% V, 99.8% EncT, 99.9% DecT} EE2-1.10-d:

			➢RA {0.00% Y, -0.01% U, 0.04% V, 99.7% EncT, 100.1% DecT } ➢LB {0.01% Y, -0.05% U, 0.10% V, 99.5% EncT, 100.3% DecT} **Decision** **• No action**
1.11	A combination of EE2-1.2c and EE2-1.6a	J. Fu, J. Zhang, S. Ma (Peking University), Y. Gao, C. Huang (ZTE), T. Dumas, K. Naser, M. Radosavljević, S. Puri (InterDigital), M.Blestel, P. Andrivon, K. Suverov, N. Zouidi, D. Ruiz Coll, J.-K. Lee (Ofinno), Y. Chang, V. Seregin, M. Karczewicz (Qualcomm Inc.)	**Summary** • Test 1.2c 와 Test 1.6a의 조합실험 • 비슷한 기고인 Test 1.12와 비교하여 trade off가 좋지 않음 • Experimental results {Y, U, V, EncT, DecT} ➢AI: Overall {-0.11%, 0.10%, 0.14%, 99.9%, 101.1%}. ➢RA: Overall {-0.xx%, -0.xx%, -0.xx%, 1xx.x%, 1xx.x%}. **Decision** **• No action**
1.12	A combination of EE2-1.2c, EE2-1.6a and EE2-1.3	J. Fu, J. Zhang, S. Ma (Peking University), Y. Gao, C. Huang (ZTE), T. Dumas, K. Naser, M. Radosavljević, S. Puri (InterDigital), M.Blestel, P. Andrivon, K. Suverov, N. Zouidi, D. Ruiz Coll, J.-K. Lee (Ofinno), Y. Chang, V. Seregin, M. Karczewicz (Qualcomm Inc.)	**Summary** • Test 1.2c 와 Test 1.6a, Test 1.3의 조합실험 • 조합실험 기고 중 가장 trade off가 좋음 • Experimental results {Y, U, V, EncT, DecT} ➢AI: {-0.15%, 0.07%, 0.09%, 100.3%, 100.3%}. ➢RA*:{-0.06%, -0.08%, 0.00%, 99.6%, 100.4%}. **Decision** **• Adopt JVET-AM0307 test 1.12**

- **Inter Prediction (3건)**

EE2	Title	Contributors	Reviews
2.1	Chained candidates in AMVP list	D. Bugdayci Sansli (Nokia)	**Summary** • AMVP / Merge list 구성 과정에 SMVP기반의 chained motion vector를 후보로 추가하는 방법이 기고됨 • Test 2.1a: Chained candidates in AMVP list • Test 2.1b: Chained candidates in AMVP and Merge list • Candidate list의 최대 후보수는 증가하지 않음 • Test 2.1b는 luma에서 작은 gain이 있으나 chroma에서 loss가 있음 • Experimental results {Y, U, V, EncT, DecT} Test 2.1a ➢RA: -0.01 %, -0.03 %, -0.02 %, 100.1%, 100.2% ➢LDB: -0.01 %, 0.10 %, 0.16 %, 100.2%, 100.1% Test 2.1b ➢RA: -0.03 %, -0.05 %, 0.01 %, 100.1%, 100.1%

			➢LDB: -0.04 %, 0.27 %, -0.07 % ,100.0%, 100.1% **Decision** **• Adopt JVET-AM0106 test 2.1b**
2.2	Regression-based GPM intra-inter prediction modification	R. Yu (Qualcomm)	**Summary** • GPM intra-inter 예측에서 intra predictor와 inter predictor를 blending할 때 blending weight를 같은 domain에서 도출하는 방법이 기고됨 • 현재 ECM은 weight 도출 과정에서 intra prediction은 LMCS domain, inter prediction은 sample domain에서 계산됨  • Test 2.2a: Blending weights derivation in original domain • Test 2.2b: Blending weights derivation in LMCS domain • 두 실험 모두 작은 gain을 가지고 있음 • Test 2.2a는 domain mapping 단계가 더 필요하지만 복잡도 차이는 미미함 • Experimental results {Y, U, V, EncT, DecT} Test 2.2a: ➢RA: -0.03% (Y), -0.03% (U), 0.03% (V); 99.9% (EncT), 100.0% (DecT) Test 2.2b: ➢RA: -0.02% (Y), -0.01% (U), 0.04% (V); 99.8% (EncT), 99.9% (DecT) **Decision** **• Adopt JVET-AM0215 test 2.2a**
2.3	Affine bilateral matching merge mode	H. Huang (Qualcomm)	**Summary** • Affine bilateral matching merge 방법이 기고됨 • DMVR조건을 충족하는 모든 병합 후보에 대해 affine BDOF 대신 픽셀단위 affine compensation을 수행함 • Encoding time이 2.7% 증가함 • BD-rate gain에 비해 encoding time이 너무 크며 trade off가 좋지 않음 • Experimental results {Y, U, V, EncT, DecT} ➢RA: -0.03%, -0.02%, -0.02% 102.7% EncT 98.9% DecT

			Decision • No action

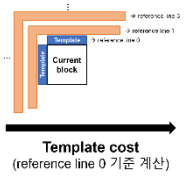
❍ **신규 기술 기고서 검토**

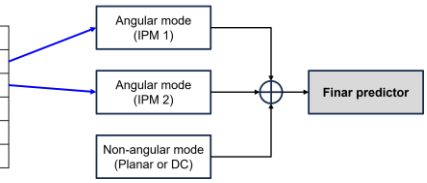
- 이번 39차 JVET 회의에서 새로운 부호화 향상 tool 제안에 대한 기술 기고서가 총 41건 (Intra 9건, Inter 9건, 기타 23건) 제출됨.
- 아래 표에 주요 기술 기고서에 관한 논의사항 및 결정사항을 요약 정리하였으며, 각 기고서의 상세한 기술적 방법은 추후 실 내 세미나를 통해서 다룰 예정임.

- **Intra Prediction (9건)**

<table>
<tr><th>Category</th><th>Title</th><th>Contributors</th><th>Reviews</th></tr>
<tr><td>Non-EE2</td><td>Non-EE2: Adaptive Planar Weight for DIMD
(JVET-AM0141)</td><td>J.-H. Lee, K. Choi (KHU),
C. W. Ryu (Kaon Group</td><td>Summary
• DIMD blending 과정에서 사용되는 planar 모드의 가중치를 적응적으로 결정하는 방법이 기고됨
• DIMD 과정을 통해 도출한 IPM의 경향을 확인한 후 3개의 IPM이 비슷한 방향성을 가질 경우 planar 모드의 가중치는 0을 부여함
• 동일한 방법으로 2개의 IPM이 비슷한 방향성을 가질 경우 가중치를 4를 부여함
• 이외의 방법은 아래 표와 같음
<table><tr><th>Condition</th><th>Planar Weight</th></tr><tr><td>3 modes in dominant range or directionally consistent</td><td>0</td></tr><tr><td>2 modes in dominant range</td><td>4</td></tr><tr><td>1 mode in dominant range</td><td>8</td></tr><tr><td>Otherwise (default case)</td><td>16</td></tr></table>• Experimental results {Y, U, V, EncT, DecT}
➢AI: 0.00%, -0.02%, -0.01%
• 아주 작은 gain을 가지고 있으며 실험 결과가 최진 ECM 버전이 아님
Decision
• No action</td></tr>
<tr><td>Non-EE2</td><td>AHG12: TMRL blend
(JVET-AM0145)</td><td>S. Blasi, G. Kulupana, D. Bugdayci Sansli, J. Lainema (Nokia)</td><td>Summary
• MRL index와 IPM 조합을 효과적으로 전송하기 위해 template cost (STAD) 기반 재정렬 및 상위 20개 조합 전송
- Multiple reference lines: 5 (1, 3, 5, 7, 12)
- Intra Prediction Modes: 10
- 총 50개 (5 * 10) 조합에 대해 template cost가 가장 낮은 20개 사용
• SATD가 가장 낮은 2개 모드와 1개 Non-angular (DC or Planar) 가중합을 통해 최</td></tr>
</table>

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<tr>
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<td>
종 예측 블록 생성하는 방법이 기고됨

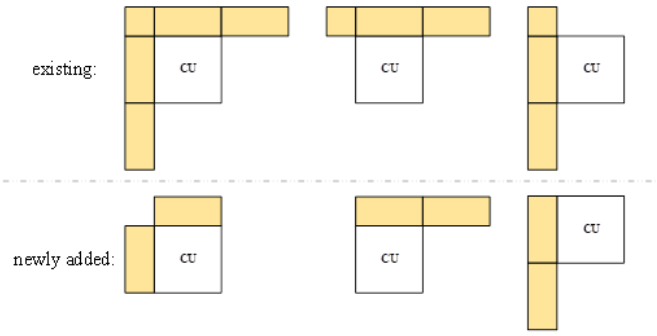
❖ Ex. TMRL list (w/o TM)
<table>
<tr><th>Index</th><th>MRL</th><th>IPM</th></tr>
<tr><td>0</td><td>1</td><td>39</td></tr>
<tr><td>1</td><td>1</td><td>38</td></tr>
<tr><td>2</td><td>1</td><td>40</td></tr>
<tr><td>3</td><td>1</td><td>37</td></tr>
<tr><td>…</td><td>…</td><td>…</td></tr>
<tr><td>49</td><td>12</td><td>5</td></tr>
</table>


❖ Ex. TMRL list
<table>
<tr><th>Index</th><th>MRL</th><th>IPM</th><th>SATD</th></tr>
<tr><td>0</td><td>3</td><td>35</td><td>80</td></tr>
<tr><td>1</td><td>1</td><td>35</td><td>90</td></tr>
<tr><td>2</td><td>1</td><td>38</td><td>100</td></tr>
<tr><td>…</td><td>…</td><td>…</td><td>…</td></tr>
<tr><td>19</td><td>12</td><td>40</td><td>700</td></tr>
</table>
❖ Ex. TMRL list
<table>
<tr><th>Index</th><th>MRL</th><th>IPM</th><th>SATD</th></tr>
<tr><td>0</td><td>3</td><td>35</td><td>80</td></tr>
<tr><td>1</td><td>1</td><td>35</td><td>90</td></tr>
<tr><td>2</td><td>1</td><td>38</td><td>100</td></tr>
<tr><td>…</td><td>…</td><td>…</td><td>…</td></tr>
<tr><td>19</td><td>12</td><td>40</td><td>700</td></tr>
</table>


• Experimental results {Y, U, V, EncT, DecT}
➢AI: -0.06%, -0.01%, -0.01%, 101.7%, 100.3%

• 추가적인 RD check로 인해 encoding time이 증가됨
• Non angular mode로 인한 gain이 얼마나 발생하는지에 대한 질문이 있음
• Encoding time 단축과 non-angular mode의 영향을 확인하기 위한 EE 진행

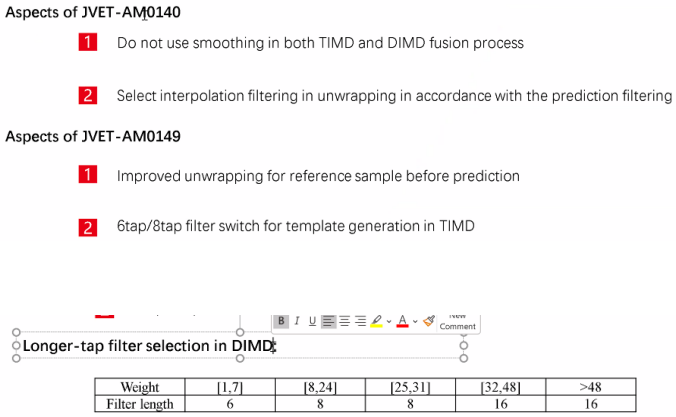
Decision
• Investigate in EE
</td>
</tr>
<tr>
<td>Non-EE2</td>
<td>Non-EE2: Extension of reconstructed area types for EIP
(JVET-AM0150)</td>
<td>W. Niu,
S. Xie,
M. Jia,
Y. Bai,
C. Huang (ZTE)</td>
<td>
Summary
• EIP를 수행하기 위한 reference 영역을 확장하는 방법이 기고됨
• 기존 3개의 영역에서 6개로 확장되었으며, 확장된 reference 영역은 아래 그림과 같음



• Experimental results {Y, U, V, EncT, DecT}
➢AI: -0.02%, -0.06%, 0.01%, 101.9%, 99.7%

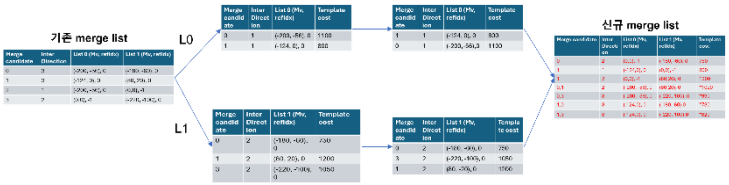
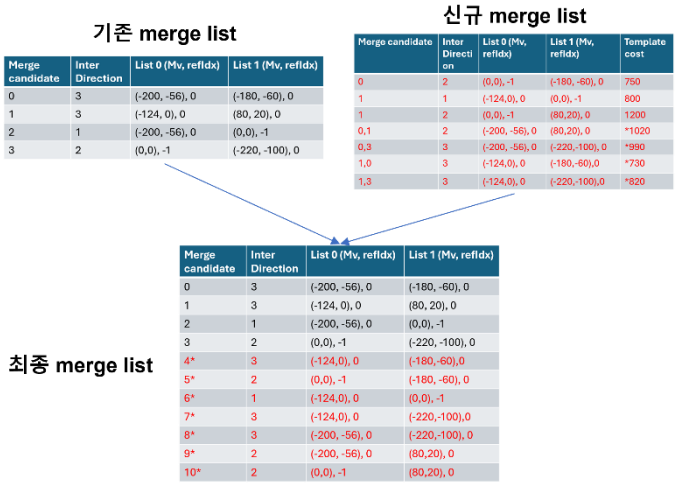
• Encoding time이 2% 가까이 상승하였음
• 그에 반해 0.02% gain은 trade off가 좋지 않음

Decision
</td>
</tr>
</table>

			• **No action**
Non-EE2	Non-EE2: On long-tap interpolation filtering for angular modes (JVET-AM0167)	T. Dong, V. Rufitskiy, A. Filippov (TCL)	**Summary** • JVET-AM0140 Non-EE2: Modifications of the unwrapping and interpolation filtering processes for angular modes 기고서와 JVET-AM0149 Non-EE2: On filtering for angular modes 를 결합한 방법이 기고됨 • 또한 DIMD fusion process에서 longer-tap interpolation filters가 사용됨  • Experimental results {Y, U, V, EncT, DecT} ➢AI: -0.06%, -0.04%, -0.04%, 99.6%, 100.2% • EE에서 아래와 같은 사항이 요구됨 • [EE2-2.2] Intra prediction smoothing • [EE2-2.3] Interpolation filter unification for unwrapping and TIMD template generation • [EE2-2.4] Longer tap interpolation filtering - DIMD predictor 생성 시 interpolation tap 개수를 weight에 따라 조정 - 최대 16-tap interpolation filter를 사용할 수 있도록 확장 • [EE2-2.5] Combination of Test 2.2, Test 2.3, and Test 2.4 **Decision** • **Investigate in EE**

- **Inter Prediction (9건)**

Category	Title	Contributors	Reviews
Non-EE2	AHG12: Generated Merge Candidates	D. Bugdayci Sansli, J. Lainema	**Summary** • Merge list를 구성하는 방법이 기고됨 • Merge list를 구성하는 방법은 아래와 같음 - 기존 merge list에 존재하는 MVP를 2개 Uni-MVP list로 분할

	(JVET-AM0107)	(Nokia)	- 각 Uni-MVP에 대해 template cost를 계산하고 정렬 - 가장 낮은 Uni-MVP들을 이용해 새로운 merge list 생성   • Experimental results {Y, U, V, EncT, DecT} ➢ RA: -0.03 % -0.08 % -0.07 %, 100.6%, 100.8% • EE에서 조사에 대한 관심이 있음 • 추가 process 감소가 요구됨 • RA의 non-LD에서만 적용되어야 함 • Worst case의 추가 작업 수 보고 필요 **Decision** **• Investigate in EE**
Non-EE2	Non-EE2: Joint reordering of GPM with intra prediction (JVET-AM0065)	Z. Sun, Y. Yu, L. Zhang, H. Yu, D. Wang (OPPO)	**Summary** • GPM의 split mode와 partition indices reordering 시에 intra prediction mode를 추가하는 방법이 기고됨 • Intra prediction, regular GPM, TM-GPM이 공동으로 reordering을 수행함 • 제안 방법 적용 여부에 대한 1ibt flag를 사용함 • Experimental results {Y, U, V, EncT, DecT} ➢ RA: -0.08%, -0.06 %, -0.11 %, EncT 101.5%, DecT 100.7% ➢ LB: -0.09%, -0.01 %, 0.06 %, EncT 101.8%, DecT 100.8%

			• Coding gain이 class 전반에 걸쳐 일관된 성능향상을 보임 • EE에서 encoding time과 성능간 trade off 개선이 요구됨 **Decision** **• Investigate in EE**
Non-EE2	Non-EE2: Joint reordering of GPM with affine prediction (JVET-AM0066)	L. Zhang, Y. Yu, Z. Sun, H. Yu, D. Wang (OPPO)	**Summary** • GPM with affine 모드의 시그널링 overhead를 줄이기 위해 affine motion information을 jointly reordering 하는 방법이 기고됨 • Candidate list의 각 후보는 1개의 split mode, 2개의 partition indices, 2개의 indicators로 구성됨 • 각 GPM partition에 대해 index는 merge list 또는 affine merge list에서 motion information을 가져오는데 사용됨 • Candidate list는 템플릿 기반 방식을 통해 reordering 됨 • Experimental results {Y, U, V, EncT, DecT} ➢RA: { -0.12% Y, -0.18% U, -0.13% V, 101.8% EncT, 101.9% DecT } ➢LDB: { -0.07% Y, 0.27% U, 0.08% V, 102.4% EncT, 101.3% DecT } • 새로운 모드에 대한 RDO process로 인해 인코딩 시간이 증가함 • EE에서 JVET-AM0065 기고서와 combination test가 요구됨 • EE에서 encoding time과 성능간 trade off 개선이 요구됨 **Decision** **• Investigate in EE**

❍ 주요 결정사항

- **[1] ECM 18.0 채택 기술**

Partioning
① N/A
Intra prediction
① [JVET-AM0307] Test 1.12: A combination of EE2-1.2c, EE2-1.6a and EE2-1.3
② [JVET-AM0163] Test 1.7a: On interpolation filter for TIMD
③ [JVET-AM0157] Test 1.8: Harmonization of SGPM-BV and LIC
④ [JVET-AM0104] EE2-related: On reference sample filtering for TIMD
Inter prediction
① [JVET-AM0106] Test 2.1b: Chained candidates in AMVP and merge lists
② [JVET-AM0215] Test 2.2a: Regression-based GPM intra-inter prediction modification
Transform and coefficient coding
① [JVET-AM0056] Test 3.1: Predictive transform coefficient coding

Loop filtering
① [JVET-AM0209] Test 4.2: Reuse of ALF control information
② [JVET-AM0063] Test 4.3: On ALF-CCCM
③ [JVET-AM0231] Test 4.5: On performance improvement of NN-ILF in ALF (non-CTC)
Entropy coding
① N/A
CTC for EE2/ECM and ECM improvements
① [JVET-AM0216] AHG12: Fixes for 12-bit internal bit depth in ECM

- **[2] 신규 EEs 구성 (JVET-AM2024)**

  - 이번 39차 JVET 회의에서 ECM tool 성능 분석을 위한 5개의 EEs가 신규 생성되었으며, 이에 대한 성능개선 결과가 차기 회의에서 논의될 예정임.

  - 총 22건의 EEs 중 Intra prediction 및 Inter prediction은 각각 8건 및 4건이 생성됨.

  - Next EE List (Intra Prediction)

EE2	Tests	Tester
2.1	TMRL blend	S. Blasi (Nokia)
2.2a 2.2b	Alignment of interpolation filters used in reference samples unwrapping and predictor generation Disabling smoothing in interpolation and reference sample processing operations for DIMD and TIMD	V. Rufitskiy (TCL)
2.3a 2.3b	Unification of unwrapping interpolation filters in TIMD template and block prediction 8-tap interpolation filter for template generation in TIMD	T. Dong (TCL)
2.4	Longer tap interpolation filtering	V. Rufitskiy (TCL)
2.5	Combination of Test 2.2, Test 2.3, and Test 2.4	V. Rufitskiy (TCL)
2.6a 2.6b 2.6c	Adaptive subsampling filter selection for CCLM/intra-CCCM/ALF-CCCM Adaptive subsampling filter selection for inter-CCCM Test 2.6a + Test 2.6b	Y. Kidani (KDDI)
2.7	Reducing candidate modes in decoder derived CCP	S. Wan (NWPU) S. Xie (ZTE)
2.8	Enhanced CCP Fusion mode	P.Bordes (InterDigital)

  - Next EE List (Inter Prediction)

EE2	Tests	Tester
3.1	Generated merge candidates	D. Bugdayci

		Sansli (Nokia)
3.2a 3.2b 3.2c	Joint reordering of GPM with affine prediction Modifications to the affine merge list Test 3.2a + Test 3.2b	L. Zhang (OPPO)
3.3	Joint reordering of GPM with intra prediction	Z. Sun (OPPO)
3.4	Test 3.2c + Test 3.3	Z. Sun (OPPO)

2.2.2. 회의결과 문서 리스트

① **JVET-AM2024** Exploration experiment on enhanced compression beyond VVC capability (EE2)

② **JVET-AM2025** Algorithm description of Enhanced Compression Model 18 (ECM 18)

3. 기존 출원 특허 및 매칭 여부 확인

생략 (공개용 보고서)

4. 향후 회의 일정

회의 차수 (MPEG / JVET)	도시	국가	일정 (MPEG 기준)
152 / 40	Geneva	CH	2025-10-07 to 2025-10-11
153 / 41	Online	-	2026-01-19 to 2026-01-23
154 / 42	Santa Eularia	ES	2026-04-27 to 2026-05-01
155 / 43	Geneva	CH	2026-07-13 to 2026-07-17
156 / 44	Hangzhou	CN	2026-10-19 o 2026-10-23

5. 주요 수집자료

생략 (공개용 보고서)